

MINIATURE AC-DC MODULE With Selectable Voltage

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The increasing use of IoT devices and smart appliances has driven demand for compact and efficient power solutions. Traditional AC-DC converters tend to be bulky, fixed-voltage, and often inefficient. This module offers a compact AC-DC solution with over 75% efficiency and selectable output, ideal for IoT systems and small appliances.

The circuit comprises an AC-DC conversion stage followed by voltage regulation and selection. The LD05-23B05SR2 (U1) buck converter converts 85-264V AC to 5V DC, which is filtered by capacitors

C1 and C2 (10 μ F each) to suppress noise. The AMS1117-3.3 (U4) then steps down the 5V to 3.3V, which is then stabilised by capacitors C3 and C4. For adjustable output, the PAM-2305DGFADJ (U5) buck converter uses inductor L1, resistors R1 (500k Ω) and R2 (120k Ω), along with capacitor C5, to provide a tunable voltage. The resistor network enables output customisation, making the module adaptable for various applications.

The AC input is connected through a 2-pin connector (U2), ensuring a secure link to the power source. DC output is routed via the WJ124-3.81-4P (J1) connector, with additional solder pads for flexible integration into IoT systems. Fig. 1 shows the module images. The Bill of Materials is listed in Table 1.

Circuit design

This circuit converts an AC input into a stable

DC output with selectable voltage options, making it well-suited for IoT devices and small appliances. It consists of three main stages: the AC-DC converter, a 3.3V regulator circuit, and a variable resistor network for adjustable output. These are all integrated with a unified output connector for streamlined distribution.

The LD05-23B05SR2 module transforms an input voltage (85V-305V AC) into a stable 5V DC output. Compact and efficient, it

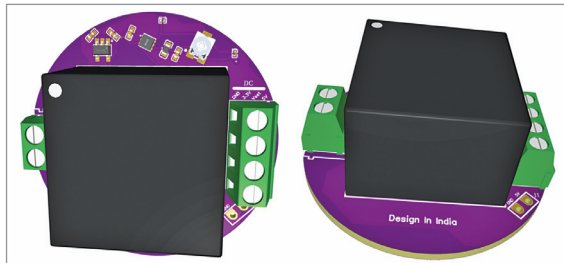


Fig. 1: Author's module

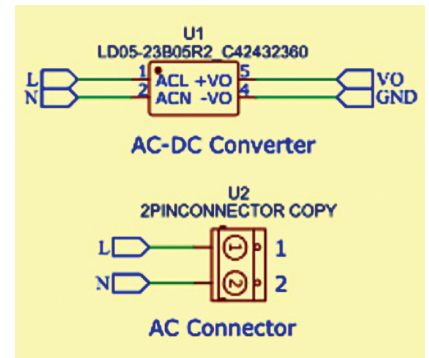


Fig. 2: Circuit diagram of AC-DC converter

TABLE 1
BILL OF MATERIALS

Name	Designator	Footprint	Quantity	Manufacturer Part	Manufacturer	Supplier Part
10 μ F	C1, C2, C3, C4	C0402	4	CL05A106MP5NUNC	Samsung	C315248
100pF	C5	C0201	1	GRM0335C1E101JA01D	Murata	C76917
4.7 μ H	L1	L0603	1	CMH160808B4R7MT	FH	C394952
WJ124-3.81-4P	P1	CONN-TH_4P-P3.81_KF124-3.81-4P	1	WJ124-3.81-4P	Kangnex	C69809
500k Ω	R1	RES-ADJ-SMD_3P-L3.8-W3.7_VGF39NCHXT	1	VGF39NCHXT-B504	HDK	C22461638
120k Ω	R2	R0201	1	0201WMF1203TEE	UNI-ROYAL	C270362
AC-DC converter	U1	PWRM-TH_LD05-23BXXR2	1	LD05-23B05R2	LZTEC	C42432360
2-pin connector	U2	2-PIN CONNECTOR	1	—	—	—
Buck converter	U3	DFN-6_L2.0-W2.0-P0.65-BL-EP	1	PAM2305DGFADJ	Diodes	C780869
3.3V regulator	U4	SOT-223	1	ME6217C33M5G	MICRONE	C427602